

ADESINA ODUNAYO BLESSING

Institute of Environment for Sustainable Development (IESD), UNEP-Tongji, University

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RESEARCH INTEREST

Throughout my career, I have been dedicated to addressing environmental challenges through innovative research and practical problem-solving. I hold a Bachelor of Science in Microbiology and a Master's degree in Environmental Science from Tongji University, China. During my Master's research, I focused on nutrient and antibiotic removal in biofilm reactors for high-ammonium wastewater treatment.

My research interests center on ecology and bioremediation for addressing pollution and climate change, including biological nutrient removal, energy recovery, and the impacts of emerging contaminants on microbial communities and ecosystem health. I am currently pursuing a PhD focused on the bioremediation of herbicide-contaminated environments, particularly the use of algal-bacterial consortia for herbicide degradation.

I aim to translate scientific advances into practical and sustainable solutions to pressing environmental and societal challenges.

EDUCATION

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| 2022-Till date | M.Sc. Environmental Science, Tongji University (UNEP-Tongji IESD)
(GPA: 4.56 out of 5.0)
Dissertation: Enhanced Nutrient and Antibiotic Removal in Algal-Bacterial Biofilm Reactors for Treating Ammonium-rich Wastewater Advisor:
Prof. Wang Rongchang |
| 2019 B.Sc. in Microbiology at the | Institute Supérieur de Communication d'Organisation et de Management (ISCOM) University, Benin Republic (GPA: 3.67 out of 5)
Dissertation: Analysis of vaginal discharge in female reproductive organs from a biological perspective.
Advisor: Allen Micheal |

MANUSCRIPTS UNDER REVIEW AND PUBLISHED PAPERS

1. William Chirwa **O.B. Adesina**, Martha Kalumbi: Ethical Issues in Lake Malawi Fisheries: A Case of Karonga District (Published by *Biodiversity - Taylor and Francis*)
<https://doi.org/10.1080/14888386.2024.2333308>)
2. **O.B. Adesina**, Chirwa William, Esther Ibiyemi Oke: Evolution in Water Treatment: Exploring Traditional Self-Purification Methods and Emerging Technologies for Drinking Water and Wastewater Treatment: A Review (Published *WNOFNS* 53 (2024) 169-185, *EISSN* 2543-5426)
3. **O.B. Adesina**, Oke Esther: Frequency and Incidence of Bacterial Infection in the Female Reproductive System (Published *WNOFNS* 52 (2024) 126-138, *EISSN* 2543-5426).
4. Eheneden Iyobosa, Rongchang Wang, Gaoxiang Chen, **O.B. Adesina**, Haijing Ren, Jie Chen, Jean Pierre Bavumiragira, Jianfu Zhao: Sulfamethoxazole removal and ammonium conversion by

- microalgae consortium: physiochemical responses and microbial community analysis. (Published Science of the Total Environment <https://doi.org/10.1016/j.scitotenv.2024.176539>)
5. A. Awomuti, Z. Yu, **O.B Adesina**, O.W. Samuel, A.W. Mumbi & D. Yin: Predictive modelling of peroxisome proliferator-activated receptor gamma (PPAR γ) IC50 inhibition by emerging pollutants using light gradient boosting machine <https://doi.org/10.1080/1062936X.2025.2478123>
 6. Adamudu Alexander Ogwuche, Iyobosa Eheneden, **Adesina O.B** Enenche Esther and Benard Obogwu: Microalgae-Based Transformation of Carbon and Nitrogen: A Pathway for Sustainable Wastewater Treatment <https://doi.org/10.9734/jgeesi/2025/v29i9943>
 7. **O.B Adesina**, J. S Onwubiko, Amoah-Nuamah John, Berha Ashenaf Alexander Ogwuche Adamudu: Textile and laundry innovations for microfiber pollution mitigation: a comprehensive review <https://doi.org/10.1007/s10452-026-10288-0>
 8. **O.B Adesina**, A.O Adamudu, J.S Onwubiko, I. Eheneden; Influencing Mechanisms in Soil Type and Associated Bacterial Community Changes in Environmental Bioremediation: A Review of Pollutant Degradation (In progress International Journal of Environmental Science and Technology)
 9. **O.B Adesina**, Shixiang Dai, Jedidiah Solomon Onwubiko, Yuqi Huang, Ying Teng: Toward Coupled Herbicide Biodegradation and Nitrogen Fixation by Algal-Bacterial Consortia in Paddy Fields: A Review of progress and future perspectives

PERSONAL ACHIEVEMENTS

1. Won Third Prize in the 2nd China-Africa Youth Innovation and Entrepreneurship Competition for designing AfricPay, a proposed cross-border payment system designed to facilitate secure, efficient, and affordable transactions between China and Africa. The system has the following key features:
 - Cutting-edge technology: Leverages blockchain, e-money payments, and artificial intelligence for secure and efficient operations.
 - Reduced costs: Aims to lower transaction fees compared to traditional methods.
 - Expanded reach: Plans to operate across the entire African continent, facilitating trade and economic connections.
 - Reduced barriers: Aims to remove hurdles in international financial transactions between China and Africa.

Link: [IESD Student Representatives Won Third Prize in the 2nd China-Africa Youth Innovation and Entrepreneurship Competition \(tongji.edu.cn\)](https://www.tongji.edu.cn/en/2024/09/20/20240920162946.htm)

2. I have reviewed research papers, for journals like the Journal of Water Process Engineering and Chemical Engineering Journal to help evaluate and enhance articles related to topics such as wastewater treatment and environmental engineering, for sustainability purposes.
3. Chinese Government Scholarship Awardee- fully funded Master's studies at Tongji University, sponsored by the China Scholarship Council.

EMPLOYMENT DETAILS

Primary Health Care Board, Abule-Oloni Ogun State
Job Title: Microbiologist

Sept 2021 - 2022

- Analyze test results for credibility and this included complete blood count (CBC), CD4 count and PCR based HIV tests for discordant results referred to the facility.
- Involved in the documentation of Covid-19 vaccine records.

Daddyclem Integrated Services

Oct 2020 - July 2021

Job Title: Quality Control/Supervisor

- I worked as a supervisor leading a direct team of bakers, sales reps, and other bakery staff.
- I worked as a quality control analyst, ensuring the right measurements went into preparation, and that the outcome met the already laid out standard.

Baptist High School Ede, Osun State. PPA(NYSC)

2019-2020

Job Title: Biology Teachers

- Effectively managed a class comprising 50% of the total student population, ensuring an engaging and inclusive learning environment.
- Offered individualized assistance to students who needed extra support, fostering an environment of personalized learning.

CERTIFICATIONS

2023	International Seminar, ASEAN-China: Asian-African Youth Dialogue, UNEP-Tongji
2023	Climate Change and Human Rights, AMNESTY INTERNATIONAL
2023	Confronting and Countering Gender-based Violence, AMNESTY INTERNATIONAL.
2023	Introduction to Amnesty International
2023	Transforming Nigerian Youths Program (By Enterprise Development Center and Young Africa Works)
2023	Course Certification by GOOGLE COURSERA (Ask Question to Make Data-Driven Decisions, Foundations: Data, Data Everywhere)
2022	Tongji University Laboratory Safety Education Training Certificate
2022	Health Safety and Environment certificate (HSE 1,2,3) from the chartered institute of Environmental Health and Safety, USA.
2020/22	Jobberman/ Accelerated Soft-skills training, (Jobberman Certificate of Achievement)

Volunteer Experience

□ International Volunteer Day Shanghai, China.	2023
□ 2 nd China-Africa Youth Innovation and Entrepreneurship competition, China.	2023
□ Immunization Program Engagement, World Health Organization (W.H.O.).	2022
□ Health Education and Advocacy Engaged Primary Healthcare Board	2022
□ Yellow Fever Vaccination Administration	2022
□ World Health Organization Measles Vaccination	2022
□ Redeem Christian Fellowship (RCF) Decoration coordinator	2019

REFERENCE

□ Available on request